```
import pandas as pd
```

```
df =pd.read_csv('/content/House Price India.csv')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14620 entries, 0 to 14619
Data columns (total 23 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   id                                    14620 non-null  int64
 1   Date                                  14620 non-null  int64
 2   number of bedrooms                    14620 non-null  int64
 3   number of bathrooms                   14620 non-null  float64
 4   living area                           14620 non-null  int64
 5   lot area                              14620 non-null  int64
 6   number of floors                      14620 non-null  float64
 7   waterfront present                    14620 non-null  int64
 8   number of views                       14620 non-null  int64
 9   condition of the house                14620 non-null  int64
 10  grade of the house                    14620 non-null  int64
 11  Area of the house(excluding basement) 14620 non-null  int64
 12  Area of the basement                  14620 non-null  int64
 13  Built Year                            14620 non-null  int64
 14  Renovation Year                       14620 non-null  int64
 15  Postal Code                           14620 non-null  int64
 16  Lattitude                             14620 non-null  float64
 17  Longitude                             14620 non-null  float64
 18  living_area_renov                     14620 non-null  int64
 19  lot_area_renov                        14620 non-null  int64
 20  Number of schools nearby              14620 non-null  int64
 21  Distance from the airport             14620 non-null  int64
 22  Price                                 14620 non-null  int64
dtypes: float64(4), int64(19)
memory usage: 2.6 MB
```

```
df.describe()
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	numbe v
count	1.462000e+04	14620.000000	14620.000000	14620.000000	14620.000000	1.462000e+04	14620.000000	14620.000000	14620.000
mean	6.762821e+09	42604.538646	3.379343	2.129583	2098.262996	1.509328e+04	1.502360	0.007661	0.233
std	6.237575e+03	67.347991	0.938719	0.769934	928.275721	3.791962e+04	0.540239	0.087193	0.766
min	6.762810e+09	42491.000000	1.000000	0.500000	370.000000	5.200000e+02	1.000000	0.000000	0.000
25%	6.762815e+09	42546.000000	3.000000	1.750000	1440.000000	5.010750e+03	1.000000	0.000000	0.000
50%	6.762821e+09	42600.000000	3.000000	2.250000	1930.000000	7.620000e+03	1.500000	0.000000	0.000
75%	6.762826e+09	42662.000000	4.000000	2.500000	2570.000000	1.080000e+04	2.000000	0.000000	0.000
max	6.762832e+09	42734.000000	33.000000	8.000000	13540.000000	1.074218e+06	3.500000	1.000000	4.000

8 rows × 23 columns

Q1. What is the average, median, and standard deviation of house prices?

```
print("Mean:", df['Price'].mean())
print("Median:", df['Price'].median())
print("Std:", df['Price'].std())
```

```
Mean: 538932.2183310534
Median: 450000.0
Std: 367532.3808039669
```

Q2. Which number of bedrooms is most common?

```
print(df['number of bedrooms'].value_counts())
```

```
number of bedrooms
3     6612
4     4724
2     1844
5     1079
6      176
```

```
1      136
7       30
8       11
9        3
10       3
33       1
11       1
Name: count, dtype: int64
```

```
df
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	number of views	condition of the house	...	Built Year	Renovation Year
0	6762810145	42491	5	2.50	3650	9050	2.0	0	4	5	...	1921	0
1	6762810635	42491	4	2.50	2920	4000	1.5	0	0	5	...	1909	0
2	6762810998	42491	5	2.75	2910	9480	1.5	0	0	3	...	1939	0
3	6762812605	42491	4	2.50	3310	42998	2.0	0	0	3	...	2001	0
4	6762812919	42491	3	2.00	2710	4500	1.5	0	0	4	...	1929	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
14615	6762830250	42734	2	1.50	1556	20000	1.0	0	0	4	...	1957	0
14616	6762830339	42734	3	2.00	1680	7000	1.5	0	0	4	...	1968	0
14617	6762830618	42734	2	1.00	1070	6120	1.0	0	0	3	...	1962	0
14618	6762830709	42734	4	1.00	1030	6621	1.0	0	0	4	...	1955	0
14619	6762831463	42734	3	1.00	900	4770	1.0	0	0	3	...	1969	2009

14620 rows × 23 columns

Q3. Check if the price data is skewed

```
print("Skewness :",df['Price'].skew())
```

```
Skewness : 4.269297720707116
```

Q4. What is the average price for each number of bedrooms?

```
print(df.groupby('number of bedrooms')['Price'].mean())
```

```
number of bedrooms
1     3.089638e+05
2     3.985476e+05
3     4.632776e+05
4     6.361988e+05
5     7.752550e+05
6     8.375815e+05
7     1.016544e+06
8     1.208455e+06
9     7.766663e+05
10    8.200000e+05
11    5.200000e+05
33    6.400000e+05
Name: Price, dtype: float64
```

Q5. What is the relationship between living area and price? (Use : Area without Basement).

```
corr = df['Area of the house(excluding basement)'].corr(df['Price'])
print("Correlation:", corr)
```

```
Correlation: 0.6152204198678108
```

Q6. Identify any anomalies where houses have high prices but low areas.

```
Start coding or generate with AI.
```

```
df
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	number of views	condition of the house	...	Built Year	Renovation Year
0	6762810145	42491	5	2.50	3650	9050	2.0	0	4	5	...	1921	0
1	6762810635	42491	4	2.50	2920	4000	1.5	0	0	5	...	1909	0
2	6762810998	42491	5	2.75	2910	9480	1.5	0	0	3	...	1939	0
3	6762812605	42491	4	2.50	3310	42998	2.0	0	0	3	...	2001	0
4	6762812919	42491	3	2.00	2710	4500	1.5	0	0	4	...	1929	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...
14615	6762830250	42734	2	1.50	1556	20000	1.0	0	0	4	...	1957	0
14616	6762830339	42734	3	2.00	1680	7000	1.5	0	0	4	...	1968	0
14617	6762830618	42734	2	1.00	1070	6120	1.0	0	0	3	...	1962	0
14618	6762830709	42734	4	1.00	1030	6621	1.0	0	0	4	...	1955	0
14619	6762831463	42734	3	1.00	900	4770	1.0	0	0	3	...	1969	2009

14620 rows × 23 columns

Q7. Compare average price based on number of floors and houses with or without waterfront

```
print(df.groupby('number of floors')['Price'].mean())

print(df.groupby('waterfront present')['Price'].mean())
```

```
number of floors
1.0    4.369769e+05
1.5    5.549301e+05
2.0    6.487372e+05
2.5    1.106233e+06
3.0    5.663281e+05
3.5    1.205875e+06
Name: Price, dtype: float64
waterfront present
0    5.304174e+05
1    1.641902e+06
Name: Price, dtype: float64
```

Q8. Identify the minimum and maximum house price. What does this indicate?

```
print('minimum :',df['Price'].min())
print('maximum :',df['Price'].max())
```

```
minimum : 78000
maximum : 7700000
```

Q9. Which location (zipcode/area) has the highest average price

```
postal_code=df.groupby('Postal Code')['Price'].mean().sort_values(ascending=False)
print(postal_code.head(5))
```

```
Postal Code
122071    2.348311e+06
122048    1.296414e+06
122057    1.188517e+06
122047    1.069295e+06
122061    8.895896e+05
Name: Price, dtype: float64
```

Q10. Write at least 5 insights from your analysis

```
# Price is heavily right-skewed
#3-bedroom homes dominate the market (45% of listings), and price rises steadily with bedroom count — from ~₹3L (1BR) up to
#Living area has only a moderate correlation with price (0.615)
#Extreme price range (₹78K to ₹77L, ~99x spread) confirms this dataset spans both budget and ultra-luxury segments, and loc
#Living area has only a moderate correlation with price (0.615)
```

```
Start coding or generate with AI.
```